



mem-test-opensuse_tinymembench

KVM testing on openSUSE Leap 16.0 via the Phoronix Test Suite.

Test Systems:

openSUSE 16.0

Processor: Intel Xeon E5-2680 v2 (8 Cores), Motherboard: QEMU Standard PC (Q35 + ICH9 2009) (4.2025.05-2 BIOS),
Memory: 1 x 16 GB RAM QEMU, Disk: 21GB QEMU HDD

OS: openSUSE Leap 16.0, Kernel: 6.12.0-160000.5-default (x86_64), Compiler: GCC 15.2.0, File-System: xfs, System
Layer: KVM

Kernel Notes: Transparent Huge Pages: always

Compiler Notes: --build=x86_64-suse-linux --disable-libcc1 --disable-libssp --disable-libstdcxx-pch --disable-libvtv --disable-plugin --disable-werror --enable-cet=auto
--enable-checking=release --enable-gnu-indirect-function --enable-host-pie --enable-host-shared --enable-languages=c,c++,objc,fortran,obj-c++,ada,go,d,jit,m2
--enable-libphobos --enable-libstdcxx-allocator=new --enable-link-serialization --enable-linux-futex --enable-multilib --enable-offload-defaulted
--enable-offload-targets=nvptx-none,amdgc-n-amdhsa, --enable-ssp --enable-version-specific-runtime-libs --host=x86_64-suse-linux --mandir=/usr/share/man
--with-arch-32=x86-64-v2 --with-arch=x86-64-v2 --with-build-config=bootstrap-lto-lean --with-gcc-major-version-only --with-libstdcxx-zoneinfo=/usr/share/zoneinfo
--with-slibdir=/lib64 --with-tune=generic --without-cuda-driver --without-system-libunwind -O2 -O2 -O2 -O2 -O2 -Werror=return-type -Werror=return-type -Werror=return-type
-Werror=return-type -fasynchronous-unwind-tables -fasynchronous-unwind-tables -fasynchronous-unwind-tables -fasynchronous-unwind-tables

-fasynchronous-unwind-tables -fstack-clash-protection -fstack-clash-protection -fstack-clash-protection -fstack-clash-protection -fstack-clash-protection -funwind-tables
 -funwind-tables -funwind-tables -funwind-tables -funwind-tables -g' -g' -g' -g'
 Processor Notes: CPU Microcode: 0x42d
 Security Notes: SELinux + gather_data_sampling: Not affected + indirect_target_selection: Not affected + itlb_multihit: Not affected + l1tf: Mitigation of PTE Inversion + mds:
 Vulnerable: Clear buffers attempted no microcode; SMT Host state unknown + meltdown: Mitigation of PTI + mmio_stale_data: Unknown: No mitigations +
 reg_file_data_sampling: Not affected + retbleed: Not affected + spec_rstack_overflow: Not affected + spec_store_bypass: Mitigation of SSB disabled via prctl + spectre_v1:
 Mitigation of usercopy/swapgs barriers and __user pointer sanitization + spectre_v2: Mitigation of Retpolines; IBPB: conditional; IBRS_FW; STIBP: disabled; RSB filling;
 PBRSE-IBRS: Not affected; BHI: Retpoline + srbds: Not affected + tsa: Not affected + tsx_async_abort: Not affected

Rocky Linux 9

Processor: Intel Xeon E5-2680 v2 (8 Cores), Motherboard: QEMU Standard PC (Q35 + ICH9 2009) (4.2025.05-2 BIOS),
 Memory: 1 x 16 GB RAM QEMU, Disk: 21GB QEMU HDD

OS: Rocky Linux 9.7, Kernel: 5.14.0-611.5.1.el9_7.x86_64 (x86_64), Compiler: GCC 11.5.0 20240719, File-System: xfs,
 System Layer: KVM

Kernel Notes: Transparent Huge Pages: always
 Compiler Notes: --build=x86_64-redhat-linux --disable-libunwind-exceptions --enable-__cxa_atexit --enable-bootstrap --enable-cet --enable-checking=release
 --enable-gnu-indirect-function --enable-gnu-unique-object --enable-host-bind-now --enable-host-pie --enable-initfini-array --enable-languages=c,c++,fortran,lto
 --enable-link-serialization=1 --enable-multilib --enable-offload-targets=nvptx-none --enable-plugin --enable-shared --enable-threads=posix --mandir=/usr/share/man
 --with-arch_32=x86-64 --with-arch_64=x86-64-v2 --with-build-config=bootstrap-lto --with-gcc-major-version-only --with-linker-hash-style=gnu --with-tune=generic
 --without-cuda-driver --without-isl
 Processor Notes: CPU Microcode: 0x42d
 Security Notes: SELinux + gather_data_sampling: Not affected + indirect_target_selection: Not affected + itlb_multihit: Not affected + l1tf: Mitigation of PTE Inversion; VMX:
 flush not necessary SMT disabled + mds: Vulnerable: Clear buffers attempted no microcode; SMT Host state unknown + meltdown: Mitigation of PTI + mmio_stale_data:
 Unknown: No mitigations + reg_file_data_sampling: Not affected + retbleed: Not affected + spec_rstack_overflow: Not affected + spec_store_bypass: Mitigation of SSB
 disabled via prctl + spectre_v1: Mitigation of usercopy/swapgs barriers and __user pointer sanitization + spectre_v2: Mitigation of Retpolines; IBPB: conditional; IBRS_FW;
 STIBP: disabled; RSB filling; PBRSE-IBRS: Not affected; BHI: Retpoline + srbds: Not affected + tsa: Not affected + tsx_async_abort: Not affected

Ubuntu 24.04

Processor: Intel Xeon E5-2680 v2 (8 Cores), Motherboard: QEMU Standard PC (Q35 + ICH9 2009) (4.2025.05-2 BIOS),
 Chipset: Intel 82G33/G31/P35/P31 + ICH9, Memory: 1 x 16 GB RAM QEMU, Disk: 21GB QEMU HDD, Audio: Intel
 82801I, Network: Red Hat Virtio device

OS: Ubuntu 24.04, Kernel: 6.8.0-101-generic (x86_64), Compiler: GCC 13.3.0, File-System: ext4, System Layer: KVM

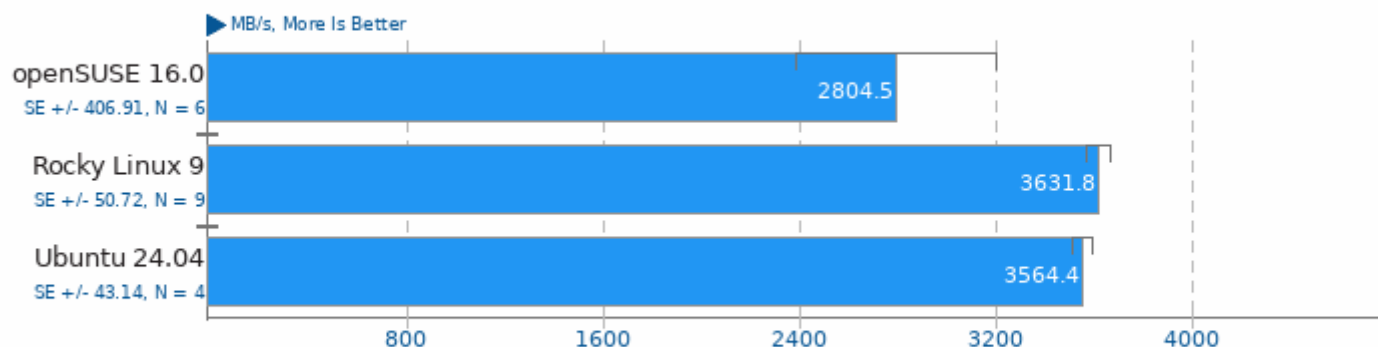
Kernel Notes: Transparent Huge Pages: madvise
 Compiler Notes: --build=x86_64-linux-gnu --disable-vtable-verify --disable-werror --enable-bootstrap --enable-cet --enable-checking=release --enable-clocale=gnu
 --enable-default-pie --enable-gnu-unique-object --enable-languages=c,ada,c++,go,d,fortran,objc,obj-c++,m2 --enable-libphobos-checking=release
 --enable-libstdcxx-backtrace --enable-libstdcxx-debug --enable-libstdcxx-time=yes --enable-link-serialization=2 --enable-multiarch --enable-multilib --enable-nls
 --enable-objc-gc=auto --enable-offload-targets=nvptx-none=/build/gcc-13-EldibY/gcc-13-13.3.0/debian/tmp-nvptx/usr,amdgc-nvptx=/build/gcc-13-EldibY/gcc-13-13.3.0/debian/tmp-gcn/usr
 --enable-plugin --enable-shared --enable-threads=posix --host=x86_64-linux-gnu --program-prefix=x86_64-linux-gnu- --target=x86_64-linux-gnu --with-abi=m64
 --with-arch-32=i686 --with-build-config=bootstrap-lto-lean --with-default-libstdcxx-abi=new --with-gcc-major-version-only --with-multilib-list=m32,m64,mx32
 --with-target-system-zlib=auto --with-tune=generic --without-cuda-driver -v
 Processor Notes: CPU Microcode: 0x42d
 Security Notes: gather_data_sampling: Not affected + indirect_target_selection: Not affected + itlb_multihit: Not affected + l1tf: Mitigation of PTE Inversion; VMX: flush not
 necessary SMT disabled + mds: Vulnerable: Clear buffers attempted no microcode; SMT Host state unknown + meltdown: Mitigation of PTI + mmio_stale_data: Unknown:
 No mitigations + reg_file_data_sampling: Not affected + retbleed: Not affected + spec_rstack_overflow: Not affected + spec_store_bypass: Mitigation of SSB disabled via
 prctl + spectre_v1: Mitigation of usercopy/swapgs barriers and __user pointer sanitization + spectre_v2: Mitigation of Retpolines; IBPB: conditional; IBRS_FW; STIBP:
 disabled; RSB filling; PBRSE-IBRS: Not affected; BHI: Retpoline + srbds: Not affected + tsa: Not affected + tsx_async_abort: Not affected + vmscape: Not affected

	openSUSE 16.0	Rocky Linux 9	Ubuntu 24.04
Tinymembench - Standard Memcpy (MB/s)	2805	3632	3564
Normalized	77.22%	100%	98.14%
Standard Deviation	35.5%	4.2%	2.4%
Tinymembench - Standard Memset (MB/s)	4368	6871	6455
Normalized	63.57%	100%	93.94%

Standard Deviation	27.1%	3.9%	2.1%
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Tinymembench 2018-05-28

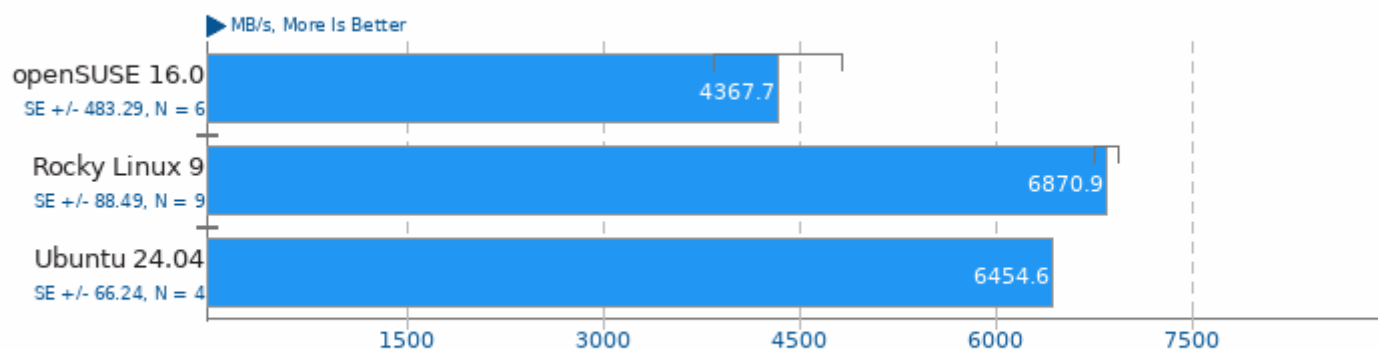
Standard Malloc



1. (CC) gcc options: -O2 -lm

Tinymembench 2018-05-28

Standard Memset



1. (CC) gcc options: -O2 -lm

This file was automatically generated via the Phoronix Test Suite benchmarking software on Friday, 13 March 2026 12:26.